535514 Reinforcement Learning

Pre-Lecture Assignment: Lecture 7

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Problem: DPG, DDPG, and Off-Policy Learning

In Lecture 6, we briefly described the motivation for learning deterministic policies. In Lecture 7, we will discuss in detail how to learn a deterministic policy by policy gradient, and this approach involves a useful RL concept called "off-policy learning." Let's try to get familiar with these concepts and get prepared for Lecture 7. You can refer to the slides **pp. 1-26** of Lecture 7 as a starting point when answering the questions below.

1) Deterministic PG: Please take a look at the DPG expression on Page 5 of the slides. Can you describe the intuition behind the expression? What's the difference between DPG and (P3) of Stochastic PG?

$$\nabla_\theta V^{\pi_\theta}(\mu) = \frac{1}{1-\gamma}\mathbb{E}_{s\sim d_\mu^{\pi_\theta}}\Big[\nabla_\theta \pi_\theta(s)\nabla_a Q^{\pi_\theta}(s,a)|_{a=\pi_\theta(s)}\Big]$$

2) Off-policy learning: Can you describe the concept of off-policy learning in 1-2 sentences? (You may refer to pp. 11-18). Moreover, can you provide a real-world example of off-policy learning?

3) Use cases of DPG: Can you guess why DPG is suitable for solving RL problems with continuous actions?

Submission Guidelines and Remarks

- Your deliverable shall be a PDF file. Please put all your write-ups into one PDF file (photos/scanned copies are acceptable) and submit your deliverable via E3.

- Please submit it by 9pm, 3/31 (Monday).

- If you have any questions, feel free to post them on Slack.